

CARIBOU ISLAND, ON, Canada

WMO: 714330

Lat: 47.339N

Lon: 85.826W

Elev: 620

StdP: 14.37

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-0.4	4.3	-3.9	4.6	0.9	1.1	5.9	6.0	41.2	24.2	38.0	25.7	14.8	10	0.831

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	8.2	70.1	65.4	67.9	64.2	66.0	62.8	66.7	68.9	64.8	67.2	63.3	65.6	9.1	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.7	97.2	68.0	63.9	90.9	66.2	62.1	85.4	64.8	31.7	68.9	30.2	67.4	29.0	65.8	72.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
36.0	31.5	28.3	-3.2	74.4	4.3	3.0	-6.3	76.5	-8.8	78.2	-11.2	79.9	-14.3	82.0
			-3.7	68.3	4.4	2.7	-6.8	70.2	-9.4	71.8	-11.8	73.3	-15.0	75.2
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
Temperatures, Degree-Days and Degree-Hours	DBAvg	40.3	21.8	22.4	27.8	35.8	42.7	48.4	55.2	60.6	56.5	46.0	36.2	28.3		
	DBStd	14.51	8.91	9.11	8.46	5.19	3.90	4.29	6.34	5.12	5.26	5.84	6.94	7.85		
	HDD50	4335	875	773	687	425	228	81	20	1	9	151	413	672		
	HDD65	9054	1340	1193	1152	875	691	498	309	152	256	588	863	1137		
	CDD50	780	0	0	0	0	2	34	182	329	205	28	0	0		
	CDD65	23	0	0	0	0	0	0	6	15	2	0	0	0		
	CDH74	5	0	0	0	0	0	0	1	4	0	0	0	0		
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0		
	WSAvg	14.3	18.1	17.7	14.6	13.9	12.3	10.6	10.0	10.9	13.2	15.9	16.7	17.9		
	PrecAvg	42.4	3.6	2.7	2.9	3.5	3.6	3.4	3.1	4.1	4.5	3.7	3.7			
Precipitation	PrecMax	53.7	6.3	5.0	6.9	5.9	5.5	7.4	6.1	5.1	8.8	7.3	5.9	6.9		
	PrecMin	33.1	1.6	0.7	0.5	1.0	1.5	1.7	1.3	1.2	1.7	2.1	1.6	0.6		
	PrecStd	5.0	1.1	1.1	1.4	1.4	1.1	1.2	1.2	1.0	1.4	1.6	1.0	1.4		
	PrecStd	5.0	1.1	1.1	1.4	1.4	1.1	1.2	1.2	1.0	1.4	1.6	1.0	1.4		
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	39.4	39.1	46.1	50.8	56.8	63.9	72.6	74.1	69.7	61.0	51.1	43.2		
		MCWB	38.3	37.9	43.4	43.7	51.1	57.5	66.3	67.9	64.7	58.4	49.4	41.6		
	2%	DB	37.5	37.3	42.0	46.3	53.0	60.0	69.2	71.0	66.8	58.4	48.6	41.3		
		MCWB	36.5	36.4	39.7	41.9	48.6	55.5	64.4	66.3	63.4	55.9	47.1	39.9		
	5%	DB	35.9	35.9	39.4	43.8	50.5	57.4	66.7	69.0	65.0	55.8	46.6	40.1		
		MCWB	35.0	35.0	37.5	40.5	47.1	53.8	62.7	65.0	62.2	53.6	45.0	38.9		
	10%	DB	33.9	34.2	37.4	41.8	48.3	55.0	64.2	67.1	63.4	54.0	45.1	38.7		
		MCWB	32.9	33.1	36.0	39.1	45.5	52.1	61.1	63.9	61.0	51.8	43.6	37.6		
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	38.8	38.4	43.6	45.8	52.8	59.0	67.6	69.3	66.6	59.3	49.9	42.2		
		MCDB	39.1	38.7	45.0	48.8	55.4	62.4	70.8	72.2	67.7	60.5	50.9	42.4		
	2%	WB	36.7	36.7	40.2	42.8	49.5	56.3	65.3	67.5	64.4	56.5	47.5	40.4		
		MCDB	37.2	37.1	41.8	45.4	51.8	59.0	68.3	69.8	66.1	58.2	48.2	40.9		
	5%	WB	35.2	35.3	37.8	41.0	47.8	54.2	63.2	66.0	62.8	54.2	45.5	39.2		
		MCDB	35.9	35.8	39.0	43.1	49.8	56.8	66.0	68.1	64.6	55.7	46.3	39.9		
	10%	WB	33.0	33.2	36.2	39.5	46.0	52.4	61.2	64.3	61.3	52.4	43.9	37.6		
		MCDB	33.8	34.0	37.2	41.4	47.7	54.7	63.8	66.7	63.1	53.9	44.9	38.7		
Mean Daily Temperature Range	5% DB	MDBR	9.6	8.8	8.6	7.7	8.9	10.1	9.6	8.2	6.8	6.0	6.6	7.8		
		MCDBR	8.4	7.1	8.7	10.5	12.6	13.8	11.5	9.6	7.9	6.8	6.1	7.1		
	5% WB	MCWBR	8.4	6.9	6.6	7.2	8.8	9.9	7.5	6.2	5.7	5.8	5.7	6.9		
		MCDBR	8.8	6.7	8.0	9.8	11.7	12.7	10.8	9.0	7.7	6.7	5.9	7.0		
	5% WB	MCWBR	9.0	6.7	6.4	7.2	9.3	10.0	7.8	6.3	6.0	5.9	5.9	7.1		
		MCDBR	9.0	6.7	6.4	7.2	9.3	10.0	7.8	6.3	6.0	5.9	5.9	7.1		
Clear-Sky Solar Irradiance	taub	0.286	0.286	0.310	0.342	0.371	0.381	0.390	0.373	0.364	0.337	0.304	0.279			
	taud	2.480	2.468	2.415	2.368	2.382	2.382	2.399	2.445	2.458	2.503	2.522	2.524			
	Ebn at Noon	258	283	291	289	283	280	275	276	268	259	248	248			
	Edh at Noon	20	25	31	36	37	37	36	33	30	24	20	17			
All-Sky Solar Radiation	RadAvg	315	590	1052	1471	1768	1910	1916	1636	1184	693	339	229			
	RadStd	36	72	130	163	134	126	108	111	75	83	50	27			

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (23 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page